

Wildlife-camera batch

Name _____

20-minute worksheet. Everything you need is on these pages. Show units in every calculation. Work alone through Parts 1 and 2. At minute 14, compare one Part 2 result with a partner, then complete Part 3 together; that comparison is included in Part 3's four minutes.

Part 1 - Clock arithmetic (about 5 minutes)

A wildlife camera's processor runs at **2.5 GHz**. Remember:

- $1 \text{ GHz} = 1,000,000,000 \text{ cycles per second}$
- $\text{cycle time} = 1 \div \text{clock rate}$
- $\text{execution time} = \text{cycles} \div \text{clock rate}$

1. Calculate the time for one processor cycle. Give the answer in seconds and nanoseconds. (1 nanosecond = 0.000000001 second.)

2. Compressing one batch requires **7.5 billion cycles**. Calculate its processor execution time at 2.5 GHz.

Part 2 - The whole batch (about 9 minutes)

Before an upgrade, one batch takes **48 minutes**:

copy and organize images: 30 minutes | processor compression: 18 minutes

Only compression changes. A new processor completes that component **3 times as fast**; the copy-and-organize work remains 30 minutes.

3. Calculate the new compression time and the new whole-batch elapsed time.

4. Calculate both speedups:

component speedup = old compression time $\div$ new compression time

whole-job speedup = old elapsed time $\div$ new elapsed time

Wildlife-camera batch

Name _____

Part 2 - The whole batch (continued)

5. Now let the processor component approach zero minutes, as if it were infinitely fast. What is the limiting elapsed time? What is the largest possible whole-job speedup from changing only this component?

6. In one sentence, explain why the component and whole-job speedups differ.

Part 3 - Read the specifications (about 4 minutes)

The replacement machine lists:

Specification	Listed value
Processor clock	maximum 3.2 GHz
Memory	24 GB capacity
Storage transfer rate	750 MB per second

For this sheet, **maximum** means a manufacturer-rated ceiling under stated conditions. **Sustained rate** means the rate observed while this job runs.

7. Label each value **rate** or **capacity**. Circle the unit evidence for each label.

Processor clock:	Memory:	Storage transfer:
------------------	---------	-------------------

8. A new image batch must be copied from storage. Can the table alone predict its copy time? Answer **yes or no**, then name the missing workload quantity.

9. The processor's **maximum 3.2 GHz** establishes what rated ceiling? Name one reason it does **not** establish the processor's sustained rate while this batch runs.

Check before the two-minute feedback. You should have two times in Part 1; new component and whole-job times, two speedups, and a limit in Part 2; and three unit-based judgments in Part 3. If you finish early, recompute question 5 by asking what remains when processor time is exactly zero. If you are stuck, write $\text{time} = \text{work} \div \text{rate}$, attach the given units, and identify which quantity is still missing.